- Welcome to Math 521!
- Syllabus is on Canvas
- 2 in-class midterms and a final exam — take note of the dates!
- Official textbook: Rudin
- Recommended reading: Ross, Copson.
- Weekly homework due on Fridays at the beginning of class on Gradescope
- Homework 1 will be posted this Friday
- Honors optional credit (and graduate coursework requirement): 1-2 additional problems for each homework. Add by next Friday!

Office hours: In the MLC, Table 3
W 9:30-10:30am
Th 2-3pm
(Starting next week)

Email: laurens@wisc.edu

- Please ask/answer any question concerning course material or logistics on Piazza.

NATURAL NUMBERS

$$\mathbb{N} = \{1, 2, 3, \dots\}$$

They satisfy the <u>Peano axioms</u>:

(N1) $1 \in \mathbb{N}$

(N2) If $n \in \mathbb{N}$ then $n+1 \in \mathbb{N}$

(N3) 1 is not equal to $n+1$ for any $n \in \mathbb{N}$

(N4) If $n, m \in \mathbb{N}$ satisfy $n+1 = m+1$, then $n = m$.

(N5) If $S \subseteq \mathbb{N}$ satifies:

- $1 \in S$
- if $n \in S$ then $n+1 \in S$

then $S = \mathbb{N}$.

Axiom (N5) forms the basis for mathematical <u>induction</u>. Suppose we want to prove that a statement $P(n)$ is true $\underline{\forall} n \in \mathbb{N}$. Then it suffices

"for all"

to verify 2 steps:

① (Base case) $P(1)$ is true.

② (Inductive step) If $P(n)$ is true for some $n \in \mathbb{N}$, then $P(n+1)$ is true.

Indeed, if we set

$$S = \{n \in \mathbb{N} : P(n) \text{ is true}\},$$

then:

- $1 \in S$: by ①
- if $n \in S$ then $n+1 \in S$: by ②

Therefore, $S = \mathbb{N}$ by axiom (N5). So $P(n)$ is

true $\forall n \in \mathbb{N}$.

<u>Ex</u> Prove

$$1^2 + 2^2 + \dots + n^2 = \frac{n(n+1)(2n+1)}{6} \qquad \forall n \in \mathbb{N}.$$

<u>Pf</u>: For $n \in \mathbb{N}$, let $P(n)$ denote

$$1^2 + 2^2 + \dots + n^2 = \frac{n(n+1)(2n+1)}{6}.$$

We will argue by induction.

① Base case: $P(1)$ is the statement $1^2 = \frac{1 \cdot 2 \cdot 3}{6}$, which is true.

② Inductive step: Assume $P(n)$ is true:

$$1^2 + 2^2 + \dots + n^2 = \frac{n(n+1)(2n+1)}{6}$$

for some $n \in \mathbb{N}$. We want to show that $P(n+1)$ is true:

$$1^2 + 2^2 + \dots + (n+1)^2 = \frac{(n+1)(n+2)(2n+3)}{6}.$$

Adding $(n+1)^2$ to both sides of $P(n)$, we get:

$$\begin{aligned}
1^2 + 2^2 + \dots + n^2 + (n+1)^2 &= \frac{n(n+1)(2n+1)}{6} + (n+1)^2 \\
&= \frac{n+1}{6}\left[n(2n+1) + 6(n+1)\right] \\
&= \frac{n+1}{6}\left[2n^2 + 7n + 6\right] \\
&= \frac{n+1}{6}(2n+3)(n+2) \\
&= \frac{(n+1)(n+2)(2n+3)}{6}
\end{aligned}$$

So $P(n+1)$ is true.

By induction, we conclude $P(n)$ is true $\forall n \in \mathbb{N}$. $\square$

<u>Ex</u> Compute the following sum for $n \in \mathbb{N}$, and prove your answer:

$$\frac{1}{1\cdot 3} + \frac{1}{3\cdot 5} + \frac{1}{5\cdot 7} + \dots + \frac{1}{(2n-1)(2n+1)}.$$

<u>Scratch work</u>:

$$\frac{1}{(2n-1)(2n+1)} = \frac{1}{2}\left(\frac{1}{2n-1} - \frac{1}{2n+1}\right)$$

$$\Rightarrow \frac{1}{1\cdot 3} + \frac{1}{3\cdot 5} + \dots + \frac{1}{(2n-1)(2n+1)}$$

$$= \frac{1}{2}\left[\frac{1}{1} - \cancel{\frac{1}{3}} + \cancel{\frac{1}{3}} - \cancel{\frac{1}{5}} + \cancel{\dots} + \cancel{\frac{1}{2n-1}} - \frac{1}{2n+1}\right]$$

$$= \frac{1}{2}\left(1 - \frac{1}{2n+1}\right) = \frac{1}{2}\cdot\frac{2n}{2n+1} = \frac{n}{2n+1}.$$

<u>Pf</u>: For $n \in \mathbb{N}$, let $P(n)$ denote the statement

$$\frac{1}{1\cdot 3} + \frac{1}{3\cdot 5} + \dots + \frac{1}{(2n-1)(2n+1)} = \frac{n}{2n+1}.$$

We will prove $P(n)$ is true $\forall n \in \mathbb{N}$ using induction.

① Base case: $P(1)$ is the statement $\frac{1}{1\cdot 3} = \frac{1}{2+1}$, which is true.

② Inductive step: Assume $P(n)$ is true:

$$\frac{1}{1\cdot 3} + \frac{1}{3\cdot 5} + \dots + \frac{1}{(2n-1)(2n+1)} = \frac{n}{2n+1}$$

for some $n \in \mathbb{N}$. Adding $\frac{1}{(2n+1)(2n+3)}$ to both sides, we get

$$\frac{1}{1\cdot 3} + \frac{1}{3\cdot 5} + \dots + \frac{1}{(2n-1)(2n+1)} + \frac{1}{(2n+1)(2n+3)} = \frac{n}{2n+1} + \frac{1}{(2n+1)(2n+3)}$$

$$= \frac{1}{2n+1}\left[n + \frac{1}{2n+3}\right] = \frac{1}{2n+1} \cdot \frac{2n^2+3n+1}{2n+3}$$

$$= \frac{\cancel{(2n+1)}(n+1)}{\cancel{(2n+1)}(2n+3)} = \frac{n+1}{2n+3}$$

So $P(n+1)$ is true.

By induction, we conclude $P(n)$ is true $\forall n \in \mathbb{N}$. $\square$

<u>Ex</u> Prove $2^n > n^2$ $\forall n \geq 5$.

n	2^n	n^2	$>$
1	2	1	T
2	4	4	F
3	8	9	F
4	16	16	F
5	32	25	T

<u>Pf</u>: For $n \in \mathbb{N}$, let $P(n)$ denote

$$2^n > n^2.$$

We will prove $P(n)$ is true $\forall n \geq 5$ using induction.

① Base case: $P(5)$ is the statement

$$32 = 2^5 > 5^2 = 25,$$

which is true.

② Inductive step: Assume $P(n)$ is true:

$$2^n > n^2$$

for some $n \geq 5$. Multiplying both sides by 2, we get

$$2^{n+1} = 2 \cdot 2^n > 2n^2 = (n+1)^2 + \underbrace{n^2 - 2n - 1}_{\geq 0?} \geq (n+1)^2$$

$$n^2 - 2n - 1 = (n-1)^2 - 2 \geq 4^2 - 2 = 14 \geq 0 \quad \text{since } n \geq 5$$

$$\Rightarrow \quad 2^{n+1} > (n+1)^2$$

So $P(n+1)$ is true.

By induction, we conclude $P(n)$ is true $\forall n \geq 5$. $\square$